

AIM:

To execute PL/SQL statements in various programs.

PL/SQL:

PL/SQL is a combination of SQL along with the procedural features of programming languages. PL/SQL is one of the key programming languages embedded in the Oracle Database along with SQL.

PROGRAM:

Set serveroutput on

1.Program to print Hello world

```
SQL> set serveroutput on
SQL> declare
  2  msg varchar(30):='Hello Yokarajan';
  3  begin
  4  dbms_output.put_line(msg);
  5  end;
  6  /
```

Hello Yokarajan

PL/SQL procedure successfully completed.

2.Program to print multiplication and division of two number:

```
SQL> declare
  2  a integer:=30;
  3  b integer:=60;
  4  c integer;
  5  d real;
  6  begin
  7  c:=a*b;
  8  dbms_output.put_line('Answer: '||c);
  9  d:=60.0/30.0;
 10  dbms_output.put_line('Answer: '||d);
 11  end;
 12  /
```

Answer:1800

Answer:2

PL/SQL procedure successfully completed.

3.Simple IF_THEN statement

```
SQL> declare
  2  num number;
  3  begin
  4  num:=&num;
```

```
5  if num>0 then
6  dbms_output.put_line('The given number is positive');
7  end if;
8  end;
9  /
```

Enter value for num: 30

old 4: num:=#

new 4: num:=30;

The given number is positive

PL/SQL procedure successfully completed.

4.Simple IF-THEN-ELSE statement

```
SQL> declare
2  num number;
3  begin
4  num:=&num;
5  if num>0 then
6  dbms_output.put_line('The given number is positive');
7  else
8  dbms_output.put_line('The given number is negative');
9  end if;
10 end;
11 /
```

Enter value for num: -30

old 4: num:=#

new 4: num:=-30;

The given number is negative

PL/SQL procedure successfully completed

5.Nested IF-THEN-ELSE

```
SQL> declare
2  num number;
3  begin
4  num:=&num;
5  if num>0 then
6  dbms_output.put_line('The given number is positive');
7  else
8  if num=0 then
9  dbms_output.put_line('The given number is equal to zero');
10 else
11 dbms_output.put_line('The given number is negative');
12 end if;
13 end if;
14 end;
15 /
```

Enter value for num: 0

old 4: num:=#

new 4: num:=0;

The given number is equal to zero

PL/SQL procedure successfully completed.

6. IF-THEN-ELSIF statement

```
SQL> declare
  2  num number;
  3  begin
  4  num:=&num;
  5  if num>0 then
  6  dbms_output.put_line('The given number is positive');
  7  elsif num=0 then
  8  dbms_output.put_line('The given number is equal to zero');
  9  else
 10  dbms_output.put_line('The given number is negative');
 11  end if;
 12  end;
 13  /
Enter value for num: 0
old   4: num:=&num;
new   4: num:=0;
The given number is equal to zero
PL/SQL procedure successfully completed.
```

7. Extended IF-THEN statement

```
SQL> declare
  2  num number:=&num;
  3  begin
  4  if num>0 then
  5  dbms_output.put_line('The given number is positive');
  6  if num>100 then
  7  dbms_output.put_line('And it is a large number');
  8  else
  9  dbms_output.put_line('And it is a small number');
 10  end if;
 11  elsif num=0 then
 12  dbms_output.put_line('The given number is equal to zero');
 13  else
 14  dbms_output.put_line('The given number is negative');
 15  end if;
 16  end;
 17  /

Enter value for num: 300
old   2: num number:=&num;
new   2: num number:=300;
The given number is positive
And it is a large number
PL/SQL procedure successfully completed.
```

8. Simple CASE statement

```
SQL> declare
  2  day_num number;
  3  day_name varchar2(10);
  4  begin
  5  day_num:=&day_num;
  6  day_name:=CASE day_num
```

```

7  when 1 then 'Sunday'
8  when 2 then 'Monday'
9  when 3 then 'Tuesday'
10 when 4 then 'Wednesday'
11 when 5 then 'Thursday'
12 when 6 then 'Friday'
13 when 7 then 'Saturday'
14 else 'Invalid day'
15 end;
16 dbms_output.put_line('Day: '||day_name);
17 end;
18 /
Enter value for day_num: 3
old   5: day_num:=&day_num;
new   5: day_num:=3;
Day:Tuesday
PL/SQL procedure successfully completed.

```

9.Searched CASE statement

```

SQL> declare
2  stock number:=&stock;
3  begin
4  case
5  when stock=0 then dbms_output.put_line('Out of stock');
6  when stock between 1 and 10 then dbms_output.put_line('Low stock');
7  when stock between 11 and 50 then dbms_output.put_line('Medium stock');
8  else dbms_output.put_line('High stock');
9  end case;
10 end;
11 /
Enter value for stock: 30
old   2: stock number:=&stock;
new   2: stock number:=30;
Low stock
PL/SQL procedure successfully completed.

```

10.EXCEPTION instead of ELSE clause in CASE statement

```

SQL> declare
2  stock number:=&stock;
3  begin
4  case
5  when stock=0 then dbms_output.put_line('Out of stock');
6  when stock between 1 and 10 then dbms_output.put_line('Low stock');
7  when stock between 11 and 50 then dbms_output.put_line('Medium stock');
8  when stock>50 then dbms_output.put_line('High stock');
9  end case;
10 exception
11     when case_not_found then dbms_output.put_line('Invalid Stock
entered');
12 end;
13 /

```

```
Enter value for stock: 0
old 2: stock number:=&stock;
new 2: stock number:=0;
Invalid Stock entered
PL/SQL procedure successfully completed.
```

11. Program for simple FOR loop

```
SQL> begin
  2  for i in 1..7
  3  loop
  4  dbms_output.put_line('Number:' || i);
  5  end loop;
  6  end;
  7  /
Number:1
Number:2
Number:3
Number:4
Number:5
Number:6
Number:7
PL/SQL procedure successfully completed.
```

12.Reverse FOR-LOOP statement

```
SQL> begin
  2  for i in reverse 1..7 loop
  3  dbms_output.put_line('Number:' || i);
  4  end loop;
  5  end;
  6  /
Number:7
Number:6
Number:5
Number:4
Number:3
Number:2
Number:1
PL/SQL procedure successfully completed.
```

13.Program using WHILE loop

```
SQL> declare
  2  i number:=3;
  3  begin
  4  while i<=7 loop
  5  dbms_output.put_line('Number:' || i);
  6  i:=i+1;
  7  end loop;
  8  end;
  9  /
Number:3
Number:4
Number:5
```

```
Number:6
Number:7
PL/SQL procedure successfully completed.
```

14.Program using DO...WHILE loop

```
SQL> declare
  2  i number:=3;
  3  begin
  4  loop
  5  dbms_output.put_line('Number:'||i);
  6  i:=i+1;
  7  exit when i > 7;
  8  end loop;
  9  end;
 10  /
```

```
Number:3
Number:4
Number:5
Number:6
Number:7
PL/SQL procedure successfully completed.
```

15.Simple GOTO statement

```
SQL> declare
  2  n number:=&n;
  3  res varchar2(10);
  4  begin
  5  if mod(n,2)=0 then
  6  res:='Even number';
  7  GOTO display_result;
  8  else
  9  res:='Odd number';
 10  GOTO display_result;
 11  end if;
 12  <<display_result>>
 13  DBMS_OUTPUT.PUT_LINE('The number ' || n || ' is an ' || res);
 14  end;
 15  /
```

```
Enter value for n: 3
old  2: n number:=&n;
new  2: n number:=5;
The number 3 is an Odd number
PL/SQL procedure successfully completed.
```

16.GOTO statement to Branch to an Enclosing Block

```
SQL> declare
  2  n number:=&n;
  3  limit number:=&limit;
  4  begin
  5  for i in n..limit loop
  6  if mod(i,5)=0 then
  7  DBMS_OUTPUT.PUT_LINE('First multiple of 5 found'|| i);
```

```

8  goto exit_label;
9  end if;
10 end loop;
11 DBMS_OUTPUT.PUT_LINE('No multiple of 5 found in the given range.');
```

12 <<exit_label>>

```

13 null;
14 end;
15 /
```

Enter value for n: 2

old 2: n number:=&n;

new 2: n number:=2;

Enter value for limit: 10

old 3: limit number:=&limit;

new 3: limit number:=10;

First multiple of 5 found5

PL/SQL procedure successfully completed.

17. FACTORIAL OF A NUMBER:

```

SQL> declare
2  n number:=7;
3  fact number:=1;
4  begin
5  for i in 1..n loop
6  fact:=fact*i;
7  end loop;
8  dbms_output.put_line('Factorial numbers are: '|| n ||' is'||fact);
9  end;
10 /
```

Factorial numbers are: 7 is5040

18.FIBBONACCI NUMBER GENERATION:

PL/SQL procedure successfully completed.

```

SQL> declare
2  v_start number:=3;
3  a number:=0;
4  b number:=1;
5  c number;
6  begin
7  dbms_output.put_line('Fibonacci series: ');
8  dbms_output.put_line(a);
9  dbms_output.put_line(b);
10 for i in 3..v_start loop
11 c:=a+b;
12 dbms_output.put_line(c);
13 a:=b;
14 b:=c;
15 end loop;
16 end;
17 /
```

Fibonacci series:

0

1

1

19.SWAPPING OF TWO NUMBERS:

PL/SQL procedure successfully completed.

```
SQL> declare
  2  n1 number :=3;
  3  n2 number :=7;
  4  temp number;
  5  begin
  6  dbms_output.put_line('Before Swapping:');
  7  dbms_output.put_line('n1:'||n1);
  8  dbms_output.put_line('n2:'||n2);
  9  temp :=n1;
 10  n1:=n2;
 11  n2:=temp;
 12  dbms_output.put_line('After Swapping:');
 13  dbms_output.put_line('n1:'||n1);
 14  dbms_output.put_line('n2:'||n2);
 15  end;
 16  /
```

Before Swapping:

n1:3

n2:7

After Swapping:

n1:7

n2:3

PL/SQL procedure successfully completed.

20.PL/SQL block for updating single row into a table

```
SQL> create table customer(cid number primary key, cname varchar2(50), city
varchar2(30), phone varchar2(15));
```

Table created.

```
SQL> declare
  2  v_cid customer.cid%type;
  3  v_cname customer.cname%type;
  4  v_city customer.city%type;
  5  v_phone customer.phone%type;
  6  begin
  7  v_cid := &v_cid;
  8  v_cname := '&v_cname';
  9  v_city := '&v_city';
 10  v_phone := '&v_phone';
 11  insert into customer values(v_cid, v_cname, v_city, v_phone);
 12  dbms_output.put_line('Customer added successfully');
```



```
13 exception
14 when others then
15 dbms_output.put_line('Error: ' || SQLERRM);
16 end;
17 /
```

```
Enter value for v_cid: 101
old 7: v_cid := &v_cid;
new 7: v_cid := 101;
Enter value for v_cname: Yokarajan
old 8: v_cname := '&v_cname';
new 8: v_cname := 'Yokarajan';
Enter value for v_city: Karur
old 9: v_city := '&v_city';
new 9: v_city := 'Karur';
Enter value for v_phone: 8838223237
old 10: v_phone := '&v_phone';
new 10: v_phone := '8838223237';
Customer added successfully
```

PL/SQL procedure successfully completed.

```
SQL> /
Enter value for v_cid: 102
old 8: v_cid := &v_cid;
new 8: v_cid := 102;
Enter value for v_cname: Viji
old 9: v_cname := '&v_cname';
new 9: v_cname := 'Viji';
Enter value for v_city: Namakal
old 10: v_city := '&v_city';
new 10: v_city := 'Namakal';
Enter value for v_phone: 9876543210
old 11: v_phone := '&v_phone';
new 11: v_phone := '9876543210';
Customer added successfully
```

PL/SQL procedure successfully completed.

```
SQL> /
Enter value for v_cid: 103
old 8: v_cid := &v_cid;
new 8: v_cid := 103;
Enter value for v_cname: Rosh
old 9: v_cname := '&v_cname';
new 9: v_cname := 'Rosh';
Enter value for v_city: Karur
old 10: v_city := '&v_city';
new 10: v_city := 'Karur';
Enter value for v_phone: 88382236339
old 11: v_phone := '&v_phone';
new 11: v_phone := '88382236339';
Customer added successfully
```

PL/SQL procedure successfully completed.

21.PL/SQL block for updating multiple rows into a table

```
SQL> declare
  2  new_name varchar2(20) := 'Yoks';
  3  begin
  4  update customer
  5  set cname = new_name where cid = 101;
  6  commit;
  7  dbms_output.put_line('Customer name updated successfully');
  8  exception
  9  when others then
 10  dbms_output.put_line('Error: ' || SQLERRM);
 11  end;
 12  /
Customer name updated successfully
```

PL/SQL procedure successfully completed.

MARK SPLITUP:

CONTENTS	MARKS ALLOTTED	MARKS OBTAINED
AIM,ALGORITHM,PL/SQL ,SQL	30	
EXECUTION AND RESULT	20	
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Result:

Thus execution of PL/SQL statements in various programs is completed.